Authentication, JWTs and try/catch

Using real world examples

Bank example

Meet Mishra ji
SBI Bank

Bank example

Meet Kirat from college
Wanting to open a bank account

Bank example

1. Kirat goes open a bank account
2. Kirat deposits some money in
3. Kirat gets back a cheque book from bank

Bank example

1. Whenever I have to pay someone, I sign the cheque
2. Everyone can see my sign
3. People can try to do the same sign on their cheque book
4. Bank won't accept theirs, will only accept my cheque

My cheque book == my json web token

Bank example

1. Something bank gives me the first time I visit it (sign in)
2. Something I need to keep safe, if I lose a cheque, someone can debit my account
3. Even though other people can see my signature/how the cheque looks, if they try to re-create it, the bank will catch them

Bank example

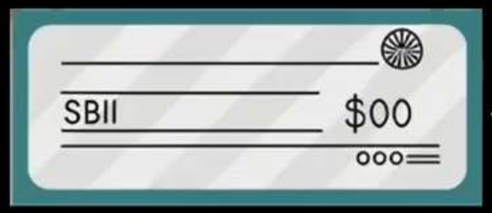
Kirat

Intruder

Bank example

Kirat

Intruder



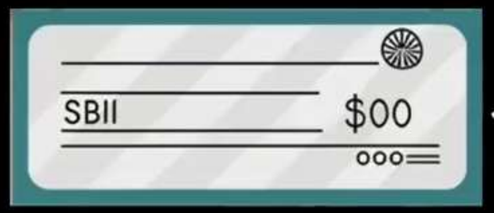
Bank example

BUT If you give them your exact signed chequebook, then they can steal your funds

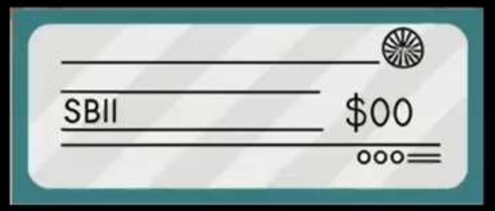
Bank example

Kirat

Intruder



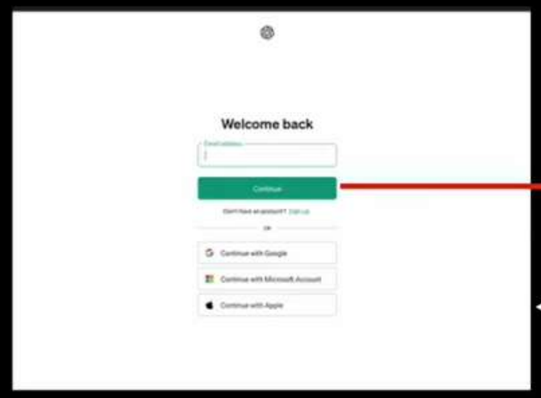
Bank example



JSON Web tokens are just like these cheques
They are issued by the backend when you sign in
Anyone can create something very similar, but the backend would reject it
If you lose the original jwt, then it is a problem

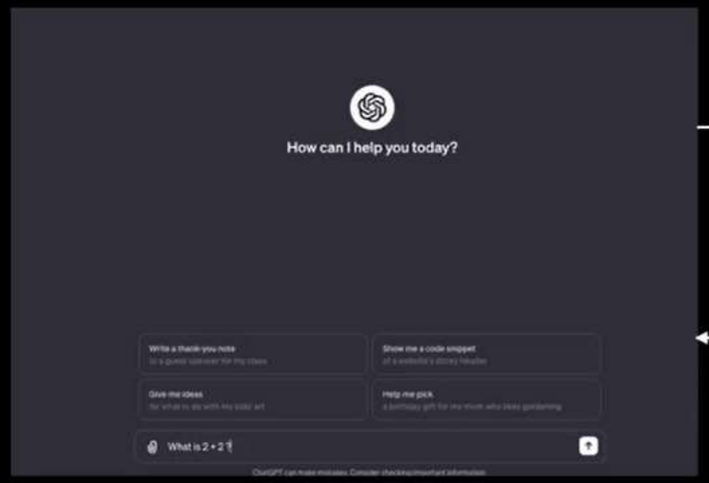
ChatGPT example

Lets look at how general signup works in a website like ChatGPT



Email, password

Jwt



Body - what is 2 + 2

Header - Jwt

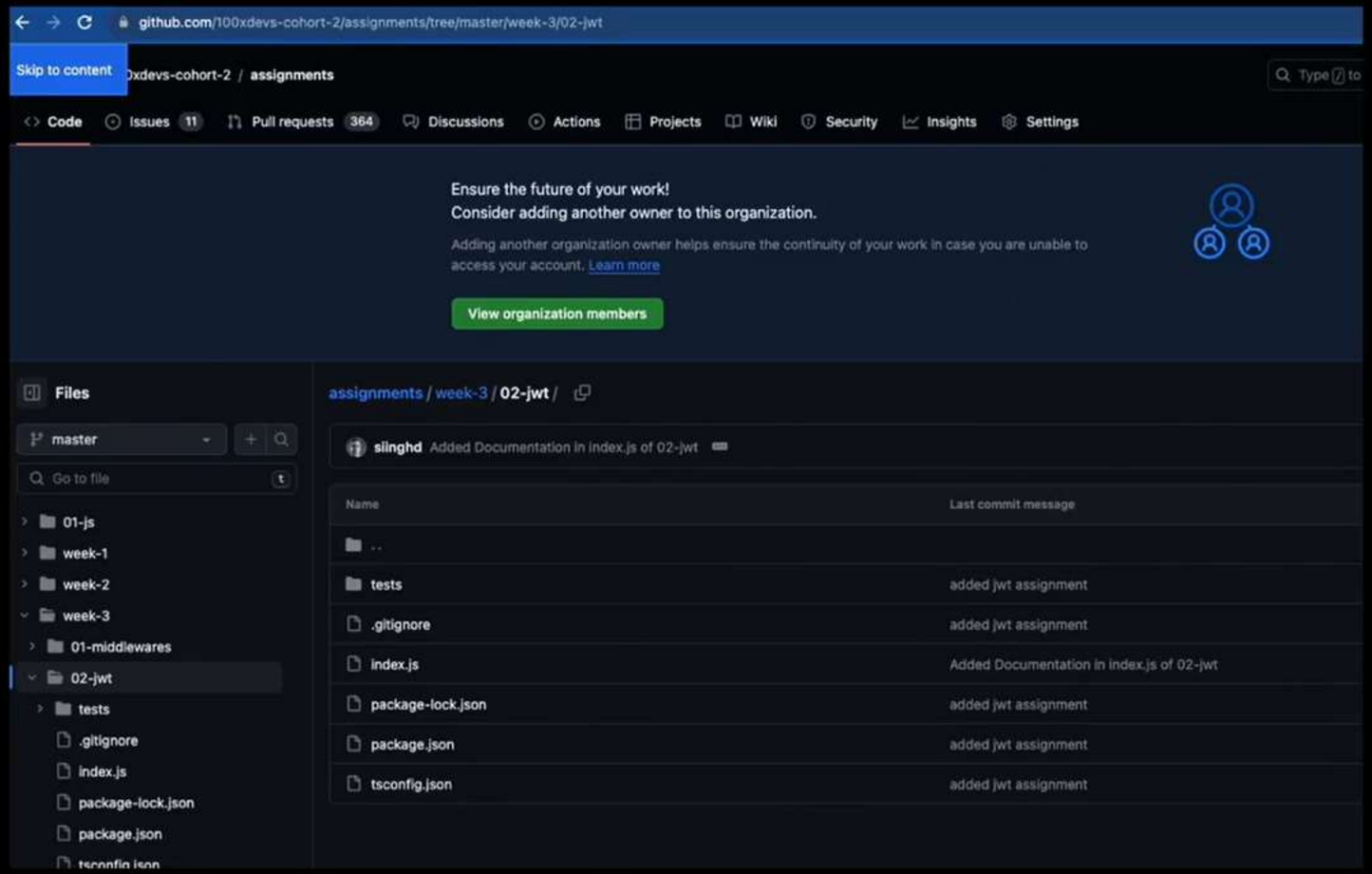
1. Verifies jwt
2. Runs model
3. Returns value

2 + 2 = 4

ChatGPT Backend

Assignment

Lets code the assignment for this week



https://github.com/100xdevs-cohort-2/assignments/tree/master/week-3/02-jwt

Try catch

Throwing and catching errors in JS

acket

Run

index.js

index.js > ...

```
1
2 function getLength(name) {
3   return name.length;
4 }
5
6 const ans = getLength("harkirat")
7 console.log(ans)
8
```

Generate

Console

Run

8

Try catch

Throwing and catching errors in JS

index.js

```
1
2  function getLength(name) {
3    return name.length;
4  }
5
6  const ans = getLength()
7  console.log(ans)
8
```

Console

```
8
```

```
/home/runner/WillingOnerlookedPacket/index.js:3
 return name.length;
            ^

TypeError: Cannot read properties of undefined (reading 'length')
    at getLength (/home/runner/WillingOnerlookedPacket/index.js:3:14)
    at Object.<anonymous> (/home/runner/WillingOnerlookedPacket/index.js:6:13)
    at Module._compile (node:internal/modules/cjs/loader:1376:14)
    at Module._extensions..js (node:internal/modules/cjs/loader:1435:10)
    at Module.load (node:internal/modules/cjs/loader:1207:32)
    at Module._load (node:internal/modules/cjs/loader:1023:12)
    at Function.executeUserEntryPoint [as runMain] (node:internal/modules/run_main:135:12)
    at node:internal/main/run_main_module:28:49

Node.js v20.10.0
```

Try catch

Throwing and catching errors in JS

index.js > ans

```
1
2 function getLength(name) {
3   return name.length;
4 }
5
6 const ans = getLength()
7 console.log(ans)
8
```

Console

Run — 355ms on 20:29:00, 12/20

```
8
```

Run — Debug with AI — 341ms on 20:29:09, 12/20

```
/home/runner/WillingOnerlookedPacket/index.js:3
 return name.length;
                ^

TypeError: Cannot read properties of undefined (reading 'length')
    at getLength (/home/runner/WillingOnerlookedPacket/index.js:3:1
4)
    at Object.<anonymous> (/home/runner/WillingOnerlookedPacket/ind
ex.js:6:13)
    at Module._compile (node:internal/modules/cjs/loader:1376:14)
    at Module._extensions..js (node:internal/modules/cjs/loader:143
5:10)
    at Module.load (node:internal/modules/cjs/loader:1207:32)
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    at node:internal/main/run_main_module:28:49

Node.js v20.10.0
```

Exception

Try catch

Throwing and catching errors in JS

When an exception is raised
the process exits since the JS program
doesn't want to proceed anymore

Control never reaches here

Packet

Run

Invite Deploy

index.js

```
1 function getLength(name) {
2   return name.length;
3 }
4
5 const ans = getLength();
6 console.log(ans);
7
8 console.log("hi there");
9
```

Console Shell

Run Debug with AI 541ms on 20:32:54, 12/20

```
/home/runner/WillingOnerlookedPacket/index.js:2
  return name.length;
                    ^

TypeError: Cannot read properties of undefined (reading 'length')
    at getLength (/home/runner/WillingOnerlookedPacket/index.js:2:1
5)
    at Object.<anonymous> (/home/runner/WillingOnerlookedPacket/ind
ex.js:5:13)
    at Module._compile (node:internal/modules/cjs/loader:1376:14)
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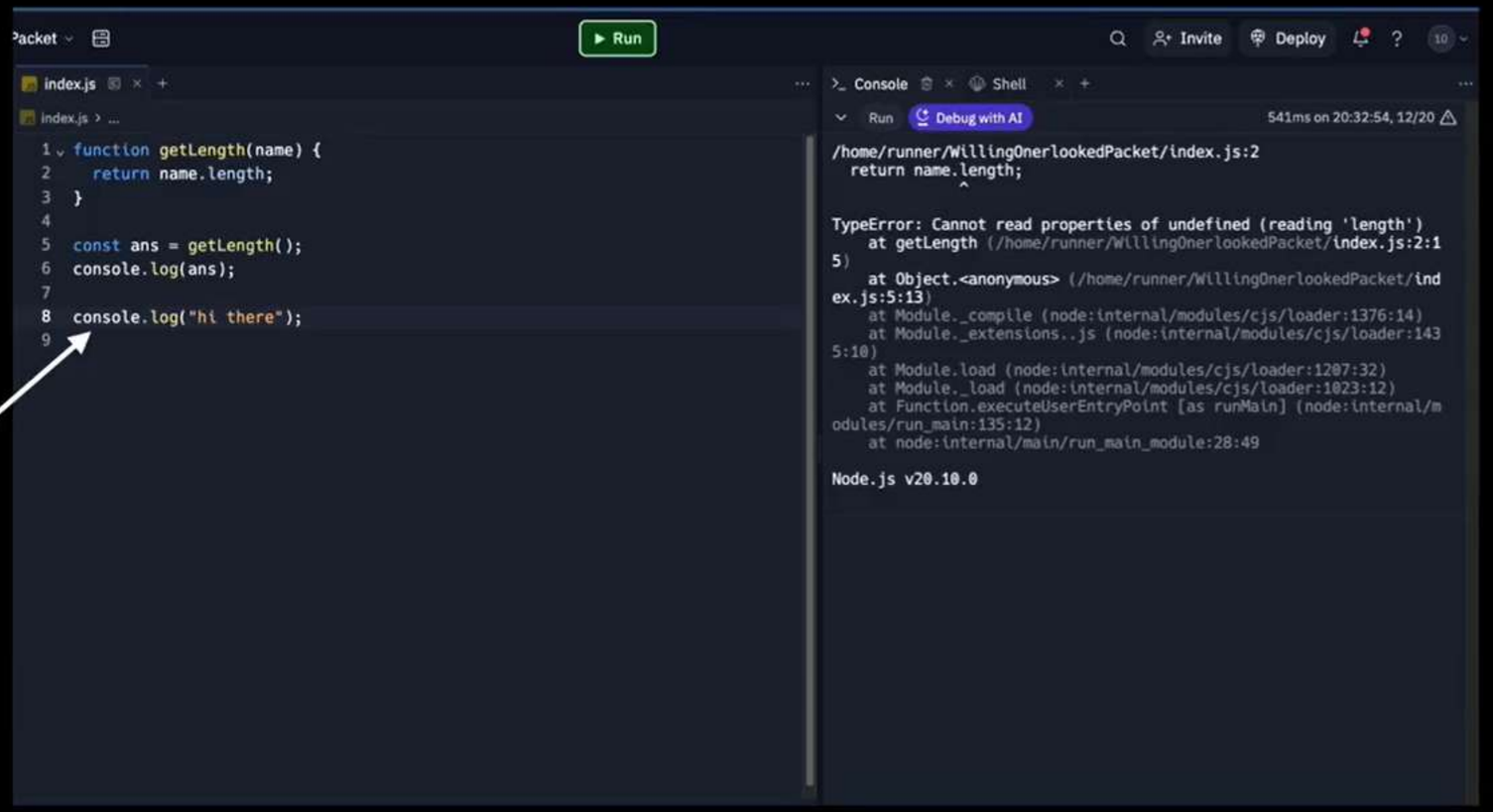
Node.js v20.10.0
```

Try catch

Throwing and catching errors in JS

But you might want the program to still continue executing. That is where you can use try catches



Try catch

Throwing and catching errors in JS

New code

index.js > ...

```
1  function getLength(name) {
2    return name.length;
3  }
4
5  try {
6    const ans = getLength();
7    console.log(ans);
8  } catch(e) {
9
10 }
11
12 console.log("hi there");
13
```

If no exception occurs,
Then it proceeds to line 12

Try catch

Throwing and catching errors in JS

New code

index.js > ...

```
1  function getLength(name) {
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If no exception occurs,
Then it proceeds to line 12

Try catch

Throwing and catching errors in JS

New code

index.js > ...

```
1  function getLength(name) {
2    return name.length;
3  }
4
5  try {
6    const ans = getLength();
7    console.log(ans);
8  } catch(e) {
9
10 }
11
12 console.log("hi there");
13
```

Normal execution proceeds after

Try catch

Throwing and catching errors in JS

New code

index.js > ...

```
1  function getLength(name) {
2    return name.length;
3  }
4
5  try {
6    const ans = getLength();
7    console.log(ans);
8  } catch(e) {
9
10 }
11
12 console.log("hi there");
13
```

If an exception occurs, then control reaches the catch